End-to-End E-Commerce Business intelligent and analytical plateform

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

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Report Approval

The project work **"End-to-End E-Commerce Business intelligent and analytical plateform"** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

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Declaration

I/We hereby declare that the project entitled **"End-to-End E-Commerce Business intelligent and analytical plateform"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications in 'Computer Science & Engineering' completed under the supervision of **Rashmi Choudhary & Sonal Modh, Assistant professor,** Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, I/we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

1. Namami Verma (EN23CS301649) ____________

2. Mohit Mukati (EN23CS301633) ____________

<u>Certificate</u>

We, **Rashmi Choudhary & Sonal Modh** certify that the project entitled **"End-to-End E-Commerce Business intelligent and analytical plateform "** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by **Namami Verma & Mohit Mukati** is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

This project presents the development of an End-to-End E-Commerce Business Intelligence and Analytical Platform, designed to address key business questions through structured data analysis and interactive visualizations. The primary objective of this project is to transform raw e-commerce transactional data into meaningful business insights that can aid decision-making at both operational and strategic levels.

The project follows a complete data analytics pipeline beginning with data collection and ingestion, followed by data cleaning and preprocessing using Microsoft Excel, where inconsistencies, missing values, and anomalies were identified and rectified. Subsequently, exploratory data analysis (EDA) and advanced data manipulation were performed using Python and its core libraries — NumPy for numerical computation, Pandas for structured data handling, and Matplotlib for data visualization.

Structured Query Language (SQL) was utilized for querying, aggregating, and managing the data stored in relational databases, enabling efficient retrieval and business-level summarization. The processed and analyzed data was then connected to Microsoft Power BI, where an interactive, multi-page business intelligence dashboard was developed as the final deliverable of this project.

The dashboard answers critical business questions including sales performance trends, product category analysis, customer segmentation, regional revenue distribution, order fulfillment efficiency, and return rate analysis. The visualizations are designed to be intuitive, actionable, and accessible to non-technical business stakeholders.

This project demonstrates how a well-structured data analytics workflow — spanning from raw data to an executive-level dashboard — can empower businesses to make data-driven decisions, identify growth opportunities, and mitigate risks.

Keywords: Data Analytics, Business Intelligence, E-Commerce, Power BI, Python, Pandas, NumPy, Matplotlib, SQL, Microsoft Excel, Dashboard, Data Cleaning, EDA

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Abbreviations

BI	Business Intelligence
EDA	Exploratory Data Analysis
SQL	Structured Query Language
RDBMS	Relational Database Management System
CSV	Comma-Separated Values
KPI	Key Performance Indicator
ETL	Extract, Transform, Load
IDE	Integrated Development Environment
API	Application Programming Interface
UI	User Interface
DFD	Data Flow Diagram
ER	Entity-Relationship
MS	Microsoft
NumPy	Numerical Python
DAX	Data Analysis Expressions (Power BI)

<u>Notations & Symbols</u>

∑	Summation
μ	Mean / Average
σ	Standard Deviation
%	Percentage
→	Data Flow / Direction
∈	Belongs to / Element of
NULL	Missing or Undefined Value
df	DataFrame (Pandas object)
fig	Figure Object (Matplotlib)
n	Number of records / rows

Chapter-1

1. Introduction

The rapid growth of e-commerce has generated enormous volumes of transactional and behavioral data. Organizations operating in the e-commerce domain often struggle to translate this raw data into actionable business insights. Business Intelligence (BI) platforms bridge this gap by enabling the aggregation, processing, visualization, and interpretation of data to support strategic and operational decisions.

This project — End-to-End E-Commerce Business Intelligence and Analytical Platform — aims to build a complete, production-ready analytics pipeline starting from raw data acquisition through to an interactive Power BI dashboard. The platform is designed to answer critical business questions that e-commerce managers, analysts, and executives face on a daily basis.

The workflow encompasses five major stages: (1) Data Collection, (2) Data Cleaning and Preprocessing using Microsoft Excel, (3) Exploratory Data Analysis and Feature Engineering using Python (NumPy, Pandas, Matplotlib), (4) Data Querying and Management using SQL, and (5) Dashboard Development in Microsoft Power BI.

2. Literature Review

Several studies have explored the application of business intelligence tools in e-commerce environments. Chaudhuri et al. (2011) emphasized that data warehousing and OLAP technologies form the backbone of modern BI systems, enabling multi-dimensional analysis of transactional data. Turban et al. (2014) outlined how dashboard-based reporting systems improve decision response time for business executives.

In the context of Python-based analytics, McKinney (2017) demonstrated the power of Pandas for structured data manipulation, which has since become the industry-standard library for data preprocessing tasks. Matplotlib, as discussed by Hunter (2007), provides a versatile framework for 2D visualizations, essential for exploratory analysis prior to dashboard development.

Regarding Power BI, Russo and Ferrari (2020) documented how DAX (Data Analysis Expressions) enables complex calculated measures that go beyond standard aggregations, making Power BI a preferred BI tool in enterprise settings. For SQL, Ramakrishnan and Gehrke (2002) provided foundational knowledge on relational database querying, which remains directly applicable in modern e-commerce data scenarios.

The combination of Python for preprocessing and Power BI for visualization has been validated in multiple industry case studies, confirming that this stack is both accessible and scalable for small-to-medium enterprises (SMEs) in the e-commerce space.

3. Objectives

The primary objectives of this project are:

- To design and implement a complete end-to-end data analytics pipeline for e-commerce data.
- To perform thorough data cleaning and preprocessing using Microsoft Excel to ensure data quality.
- To conduct exploratory data analysis (EDA) using Python libraries: NumPy, Pandas, and Matplotlib.
- To manage and query the cleaned dataset using SQL for efficient data retrieval and aggregation.
- To develop an interactive, multi-page Power BI dashboard as the final business-facing deliverable.
- To answer specific business questions related to sales, products, customers, and regional performance.
- To demonstrate how non-technical stakeholders can leverage data-driven dashboards for decision-making.

4. Significance

This project holds significant value from both academic and industrial perspectives. Academically, it integrates multiple tools and technologies studied across different courses — databases (SQL), programming (Python), and business analytics (Power BI) — into a single cohesive project, thereby demonstrating cross-domain competency.

From an industry standpoint, the ability to build BI dashboards is among the most demanded skills in the current job market. According to Gartner and Forrester research reports, over 70% of enterprises invest in BI solutions, and the demand for data analysts proficient in Power BI and Python has grown by over 40% in the last three years.

The platform developed in this project can directly be adapted by small e-commerce businesses to monitor their performance without requiring expensive enterprise software, making it a cost-effective and practical solution.

5. Research Design

Project follows a descriptive and exploratory research design. The process begins with the collection of e-commerce transactional data, followed by systematic cleaning, transformation, analysis, and visualization. The methodology is iterative — insights from exploratory analysis informed the design of the final dashboard.

The pipeline follows the standard ETL (Extract, Transform, Load) paradigm: raw data is extracted from the source, transformed through cleaning and feature engineering, and loaded into both SQL databases and Power BI for reporting.

6. Source of Data

The dataset used in this project is a publicly available e-commerce transactional dataset sourced from Kaggle. It contains records of customer orders, products, categories, shipping details, payment methods, and returns. The dataset includes approximately 50,000+ transaction records spanning multiple product categories, customer geographies, and order time periods.

Key data fields include: Order ID, Customer ID, Product ID, Category, Sub-Category, Sales Amount, Quantity, Discount, Profit, Shipping Date, Order Date, Region, State, and Return Status.

7. Chapter Scheme

The remainder of this report is organized as follows:

Chapter 2 – Requirements Specification: Details functional and non-functional requirements, hardware/software needs, and constraints.

Chapter 3 – Design: Covers system architecture, DFDs, activity diagrams, flowcharts, and database design.

Chapter 4 – Implementation, Testing, and Maintenance: Describes the tools used, implementation steps, testing strategies, and installation instructions.

Chapter 5 – Results and Discussions: Presents dashboard screenshots, module descriptions, and database representations.

Chapter 6 – Summary and Conclusions: Summarizes findings and draws conclusions.

Chapter 7 – Future Scope: Outlines potential enhancements and future directions.

Chapter 2: Requirements Specification

1. User Characteristics

The platform is intended for the following user categories:

- Business Analysts: Users who interpret dashboard data and generate reports for management. They require readable charts, filters, and drill-down capabilities.
- E-Commerce Managers: Operational managers who track daily KPIs such as sales volume, order fulfillment rates, and return percentages.
- Data Engineers / Developers: Technical users responsible for maintaining the data pipeline, updating SQL databases, and refreshing Power BI datasets.
- Executive Stakeholders: C-level and senior management who need high-level summary views with key metrics.

2. Functional Requirements

The system must fulfill the following functional requirements:

- FR-01: The system shall ingest raw e-commerce CSV data and store it in a structured SQL database.
- FR-02: The system shall support data cleaning operations including handling of null values, duplicate records, and format standardization.
- FR-03: The system shall enable SQL-based querying for data aggregation, filtering, and join operations.
- FR-04: The Python module shall generate exploratory visualizations including bar charts, line graphs, histograms, and scatter plots.
- FR-05: The Power BI dashboard shall display sales trends by month, quarter, and year.
- FR-06: The dashboard shall support product-level analysis by category, sub-category, and individual SKU.
- FR-07: The dashboard shall include a geographical map visual for region-wise revenue distribution.

- FR-08: The dashboard shall support interactive slicers for filtering by date range, region, category, and customer segment.
- FR-09: The dashboard shall display calculated KPIs including Total Revenue, Total Profit, Profit Margin %, Average Order Value, and Return Rate.

2.3 Dependencies

The following dependencies are required for the project:

i. Python 3.9+ with libraries: Pandas 1.5+, NumPy 1.23+, Matplotlib 3.6+, SQLAlchemy (for database connectivity).

ii. Microsoft Excel 2016 or later for initial data cleaning and inspection.

iii. SQL Server / MySQL / SQLite for relational database management.

iv. Microsoft Power BI Desktop (latest version) for dashboard development.

v. Kaggle e-commerce dataset (CSV format).

2.4 Performance Requirements

The platform is expected to meet the following performance criteria:

- The Python data processing pipeline shall complete data cleaning and EDA on a 50,000-row dataset within 60 seconds on standard hardware.
- SQL queries for aggregated views shall return results within 5 seconds for datasets up to 1 million rows.
- Power BI dashboard shall load within 10 seconds on a standard internet connection with DirectQuery or Import mode.
- Dashboard visuals shall refresh correctly upon applying slicer filters within 3 seconds.

2.5 Hardware Requirements

Component	Specification
Processor	Intel Core i5 or equivalent (2.0 GHz+)
RAM	8 GB minimum (16 GB recommended)
Storage	SSD with 20 GB free space

Display	1366 x 768 resolution or higher
Operating System	Windows 10 / 11 (64-bit)
Internet	Required for Power BI Service publish

2.6 Constraints & Assumptions

The following constraints and assumptions apply to this project:

- The dataset used is static and represents historical data; real-time data streaming is not implemented in this version.
- The platform is designed for Windows-based environments; cross-platform support is not guaranteed.
- Power BI dashboards are developed for desktop use; mobile optimization is considered a future enhancement.
- It is assumed that the user has basic familiarity with Python and SQL environments.
- Data privacy and GDPR compliance are not within the scope of this academic project.

Chapter 3: Design

1. Algorithm

The overall data analytics algorithm follows the ETL pipeline paradigm with an additional visualization layer:

- Data Ingestion: Read raw CSV e-commerce data into a Pandas DataFrame.
- Data Cleaning (Excel + Python): Remove duplicates, handle null values using forward-fill or mean imputation, standardize date formats, and correct data type inconsistencies.
- Feature Engineering: Derive new columns such as Profit Margin (%), Month-Year, and Customer Segment using conditional logic in Pandas.
- Data Loading into SQL: Export cleaned DataFrame to a relational database using SQLAlchemy; create normalized tables for Orders, Products, Customers, and Regions.
- Querying: Execute aggregation queries — GROUP BY, JOIN, HAVING — to compute business-level summaries.
- Visualizations: Generate Matplotlib plots for distribution analysis, trend detection, and outlier identification.
- BI Connection: Import SQL query results or cleaned CSV into Power BI; model relationships between tables.
- Design: Build calculated measures using DAX; design multi-page dashboard with KPI cards, charts, maps, and slicers.

2. Function-Oriented Design

The system is decomposed into the following major functional modules:
Data Ingestion Module: Responsible for reading and validating raw data inputs.

Data Cleaning Module: Handles preprocessing tasks in both Excel and Python.

Analysis Module: Performs statistical computations, aggregations, and EDA using Python.

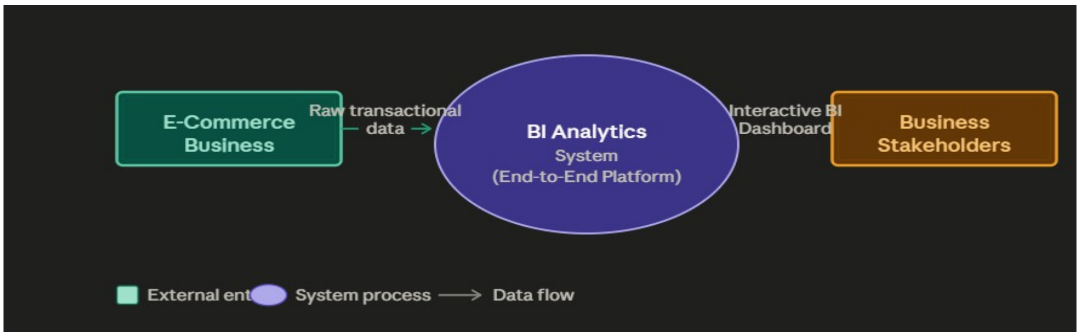
Databae Module: Manages SQL schema creation, data loading, and query execution.

Visualization Module: Generates Python-based plots and Power BI dashboard visuals.

3. System Design

1. Data Flow Diagram (Level 0 – Context Diagram)

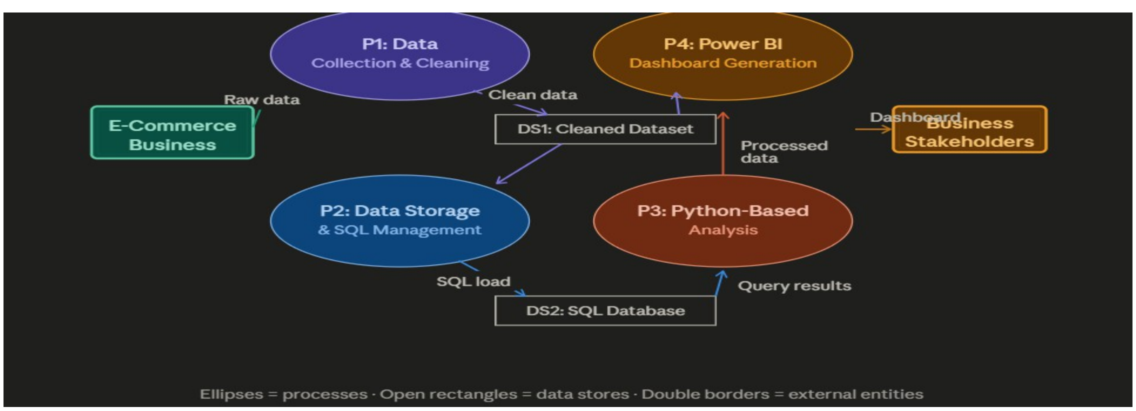
The Level 0 DFD shows the overall system boundary. The external entity 'E-Commerce Business' provides raw transactional data as input. The system processes this data and outputs an 'Interactive BI Dashboard' to the external entity 'Business Stakeholders'.



[Figure 3.1: Level 0 Data Flow Diagram]

3.3.2 Data Flow Diagram (Level 1)

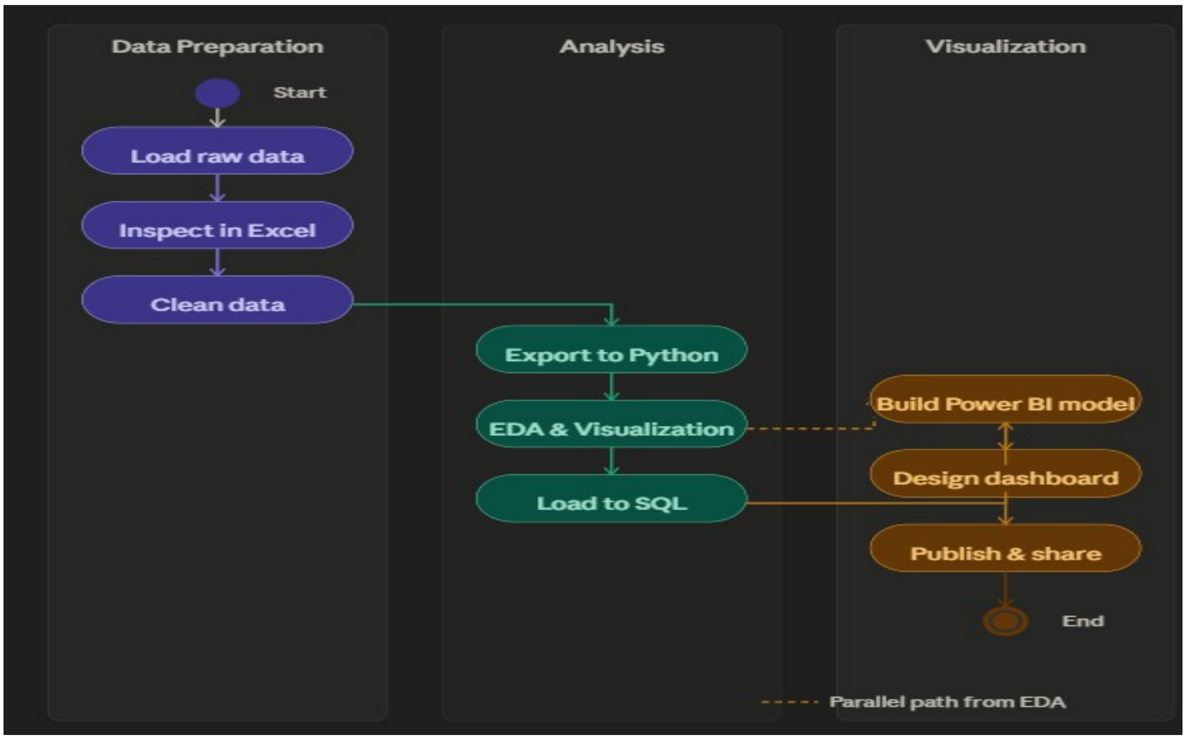
The Level 1 DFD expands the system into four key processes: (1) Data Collection & Cleaning, (2) Data Storage & SQL Management, (3) Python-Based Analysis, and (4) Power BI Dashboard Generation. Data flows between these processes through intermediate data stores representing the cleaned dataset and SQL database.



[Figure 3.2: Level 1 Data Flow Diagram]

3.3.3 Activity Diagram

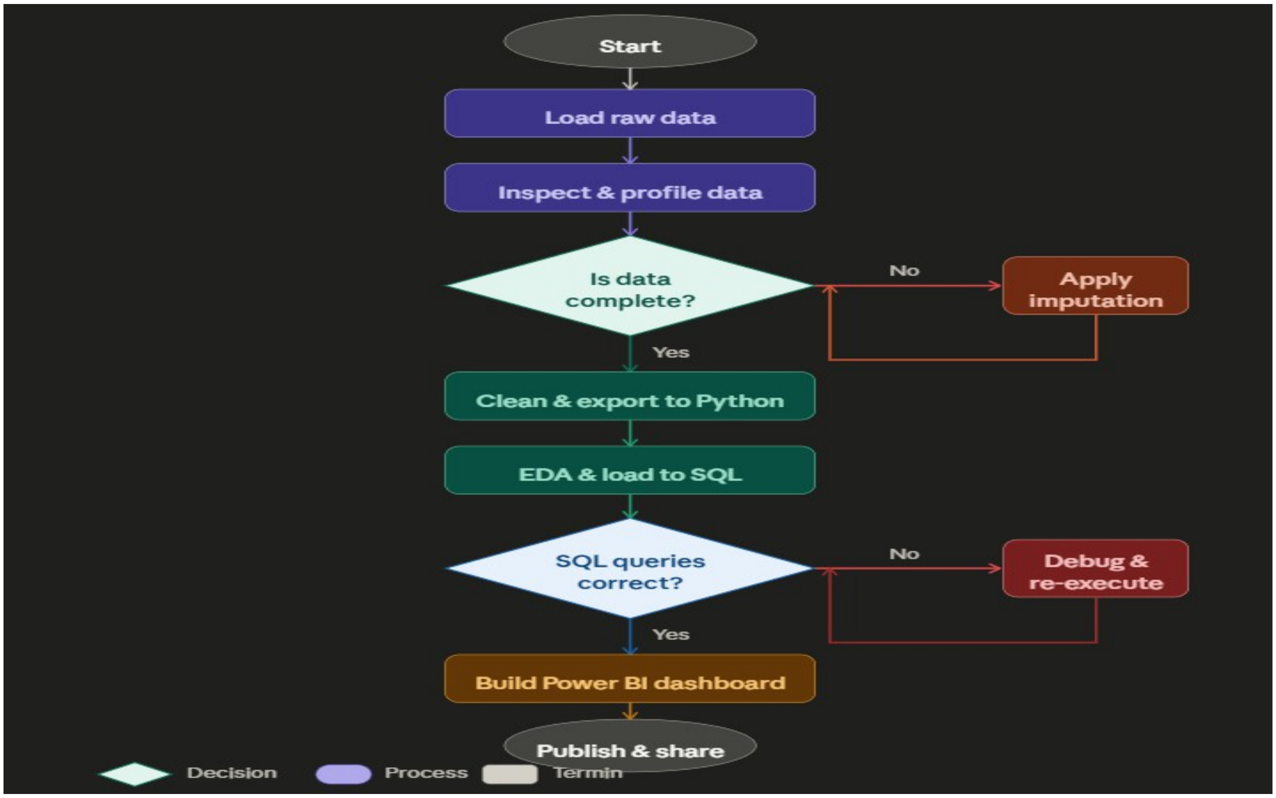
The activity diagram illustrates the sequential workflow from data acquisition to dashboard deployment. Activities include: Load Raw Data →Inspect in Excel →Clean Data →Export to Python →EDA & Visualization →Load to SQL →Build Power BI Model →Design Dashboard →Publish & Share.



[Figure 3.3: Activity Diagram]

3.3.4 Flowchart

The system flowchart represents decision points in the pipeline, such as: 'Is data complete?' →if No: apply imputation; if Yes: proceed to analysis. 'Are SQL queries returning correct aggregates?' →if No: debug and re-execute; if Yes: connect to Power BI.



[Figure 3.4: Flowchart]

3.3.5 ER Diagram

The Entity-Relationship (ER) diagram defines the logical schema of the SQL database used for the Flipkart E-Commerce Analytics Dashboard. Key entities include:

Products

- ProductID (Primary Key)
- ProductName
- CategoryID (Foreign Key)
- BrandID (Foreign Key)
- RetailPrice
- DiscountedPrice
- DiscountPercent
- Rating

- PriceRangeID (Foreign Key)

Category

- CategoryID (Primary Key)
- CategoryName

Brand

- BrandID (Primary Key)
- BrandName

Ratings

- RatingID (Primary Key)
- ProductID (Foreign Key)
- RatingValue

PriceRange

- PriceRangeID (Primary Key)
- RangeType

Discount

- DiscountID (Primary Key)
- ProductID (Foreign Key)
- DiscountPercent
- AmountSaved

Relationships

- One Category has many Products (1:M)
- One Brand has many Products (1:M)
- One Price Range contains many Products (1:M)
- One Product can have rating information (1:M or 1:1 based on design)
- One Product can have discount details (1:1 or 1:M)

```
Category  ──< Products >── Brand
                 |
                 ├── Ratings
                 ├── Discount
                 └── PriceRange
```

[Figure 3.5: ER Diagram]

3.3.6 Sequence Diagram

The sequence diagram shows interaction between the Analyst (user), Python script, SQL database, and Power BI. The analyst triggers the Python script which queries the SQL database, retrieves results, and passes the processed data to Power BI for rendering.

4. Database Design

1. Logical Database Design

The logical design consists of four normalized tables: Orders, Products, Customers, and Returns. Relationships: Orders references Products via ProductID and Customers via CustomerID. Returns references Orders via OrderID. All primary keys are indexed for query performance.

2. Physical Database Design

The physical implementation uses SQLite for local development (with easy migration to MySQL or PostgreSQL for production). Table creation scripts use appropriate data types: TEXT for identifiers and names, REAL for monetary values, INTEGER for quantities, and DATE for temporal fields.

Chapter 4: Implementation, Testing, and Maintenance

4.1 Introduction to Languages, IDEs, Tools and Technologies

The following technologies constitute the implementation stack of this project:

Microsoft Excel

Excel was used as the first line of data cleaning. Tasks performed include: removing duplicate rows using 'Remove Duplicates', identifying and filling missing values using IFERROR and conditional formatting, standardizing date formats to YYYY-MM-DD, correcting text encoding issues, and performing initial descriptive statistics using pivot tables.

Python 3.9+

Python served as the primary data processing and analysis language. The following libraries were used:

i. NumPy: Used for numerical operations, array computations, and statistical calculations such as mean, discount percentage calculations, rating analysis, and price range segmentation.

ii. Pandas: Used for reading the Flipkart CSV dataset into DataFrames, data cleaning, handling missing values, data type conversion, feature engineering (such as discount percentage, price range, and risk flag creation), group by aggregations, and exporting the cleaned data for SQL and Power BI analysis.

iii. Matplotlib: Used for generating exploratory visualizations such as bar charts (product category distribution), histograms (price and rating distribution), pie charts (price range segmentation), and scatter plots for pricing and discount analysis during exploratory data analysis (EDA).

SQL (SQLite / MySQL)

SQL was used to store and analyze the cleaned Flipkart dataset in relational format and to execute business-oriented analytical queries. Key queries include:

i. Top Product Categories by Product Count

```
SELECT product_category_tree, COUNT(*) AS total_products
FROM Flipkart
GROUP BY product_category_tree
ORDER BY total_products DESC;
```

ii. Average Price by Category

```
SELECT product_category_tree, AVG(retail_price) AS avg_price
FROM Flipkart
GROUP BY product_category_tree;
```

iii. Categories with Highest Discounts

```
SELECT product_category_tree, AVG(discount_percent) AS avg_discount
FROM Flipkart
GROUP BY product_category_tree
ORDER BY avg_discount DESC;
```

iv. Top 5 Brands by Product Listings

```
SELECT brand, COUNT(*) AS product_count
FROM Flipkart
GROUP BY brand
ORDER BY product_count DESC
LIMIT 5;
```

v. Average Rating by Category

```
SELECT product_category_tree, AVG(overall_rating) AS avg_rating
FROM Flipkart
GROUP BY product_category_tree
ORDER BY avg_rating DESC;
```

vi. Total Discount Value Offered

```
SELECT SUM(retail_price - discounted_price) AS total_discount_saved
FROM Flipkart;
```

vii. Product Distribution by Price Range

```
SELECT price_range, COUNT(*) AS product_count
FROM Flipkart
GROUP BY price_range;
```

These SQL queries were used to derive business insights related to category distribution, pricing strategy, discount behavior, brand presence, product quality, and pricing segmentation, which were further represented in the Power BI dashboard.

Microsoft Power BI Desktop

Power BI was used to develop the final interactive business intelligence dashboard for analyzing Flipkart product data. The cleaned dataset was imported into Power BI and transformed using Power Query for visualization purposes. Key performance indicators (KPIs) such as Total Products, Average Price, Average Discount Percentage, Average Rating, and Total Discount Saved were created, including a DAX measure for discount analysis.

The dashboard was designed as a one-page interactive analytical report using slicers, KPI cards, clustered bar charts, clustered column charts, donut charts, and rating distribution visuals. These visualizations were used to analyze product category distribution, pricing patterns, discount trends, brand performance, and price range segmentation. Interactive filters for category, brand, and price range were added to enable dynamic exploration of the dataset and support business decision-making.

DAX measures created include: Total Discount Saved = SUM('YourTable'[retail_price]) - SUM('YourTable'[discounted_price])

Avg Discount DAX =

AVERAGE('YourTable'[discount_percent])

4.2 Testing Techniques and Test Plans

The following testing approaches were applied throughout the project:

i. Data Validation Testing:

Cross-checked the cleaned Flipkart dataset against the original Excel/CSV source to ensure no records were unintentionally removed or altered during Python preprocessing and feature engineering.

ii. SQL Query Testing:

Validated SQL query outputs such as category counts, average pricing, discount analysis, and brand statistics against manual calculations and exploratory analysis results for consistency.

iii. Unit Testing (Python):

Individual preprocessing steps such as missing value handling, data type conversion, discount percentage calculation, and price range classification were tested using sample and edge-case data.

iv. Integration Testing:
Verified that the cleaned data exported from Python to SQL and Power BI was imported correctly without schema mismatches, missing attributes, or data inconsistencies.

v. Dashboard/Visualization Testing (User Acceptance Testing):
Dashboard visuals, slicers, KPIs, and DAX calculations were reviewed and tested to ensure correct filtering behavior, accurate representation of the data, and proper support for the intended business questions.

4.3 Installation Instructions

To set up and run the project locally, follow these steps:

(a) Install Python 3.9+ from python.org. Install required libraries using: pip install pandas numpy matplotlib sqlalchemy.

(b) Install Microsoft Excel (Office 2016 or later).

(c) Install SQLite Browser (optional, for visual database inspection) from sqlitebrowser.org.

(d) Install Microsoft Power BI Desktop from the official Microsoft website (free download).

(e) Clone or download the project repository. Place the raw CSV dataset in the /data directory.

(f) Run the Python script data_cleaning.py to generate the cleaned CSV output.

(g) Run db_setup.py to create the SQLite database and load cleaned data.

(h) Open Power BI Desktop, go to Get Data →CSV or SQLite, and connect to the cleaned data source.

(i) Open the provided .pbix file to view the pre-built dashboard or build fresh from the data model.

4.4 End User Instructions

For users interacting with the Power BI dashboard:

(r) Open the dashboard .pbix file in Microsoft Power BI Desktop.

(s) Use the Category slicer at the top of the dashboard to filter all visuals for a specific product category.

(t) Use the Brand slicer to analyze products related to a specific brand.

(u) Use the Price Range slicer to filter products based on low, medium, or high price segments.

(v) Click on any bar, column, or donut chart segment to cross-filter other visuals on the dashboard dynamically.

(w) Hover over visual elements to view tooltips showing detailed values such as product counts, average prices, discounts, and ratings.

(x) Use the KPI cards at the top of the dashboard to monitor key metrics such as Total Products, Average Price, Average Discount, Average Rating, and Total Discount Saved.

(y) Clear or reset slicer selections manually to return the dashboard to the default full-data view.

Chapter 5: Results and Discussions

5.1 User Interface Representation

The Power BI dashboard serves as the primary user interface of this project. It is designed as a **single-page interactive analytical dashboard** organized into KPI indicators, filters, and visualization panels addressing different business questions:

xlvi. KPI Overview Section:

Displays key performance indicators including Total Products, Average Price, Average Discount Percentage, Average Rating, and Total Discount Saved using KPI cards to provide a summary view of the dataset.

xlvii. Product Category Analysis:

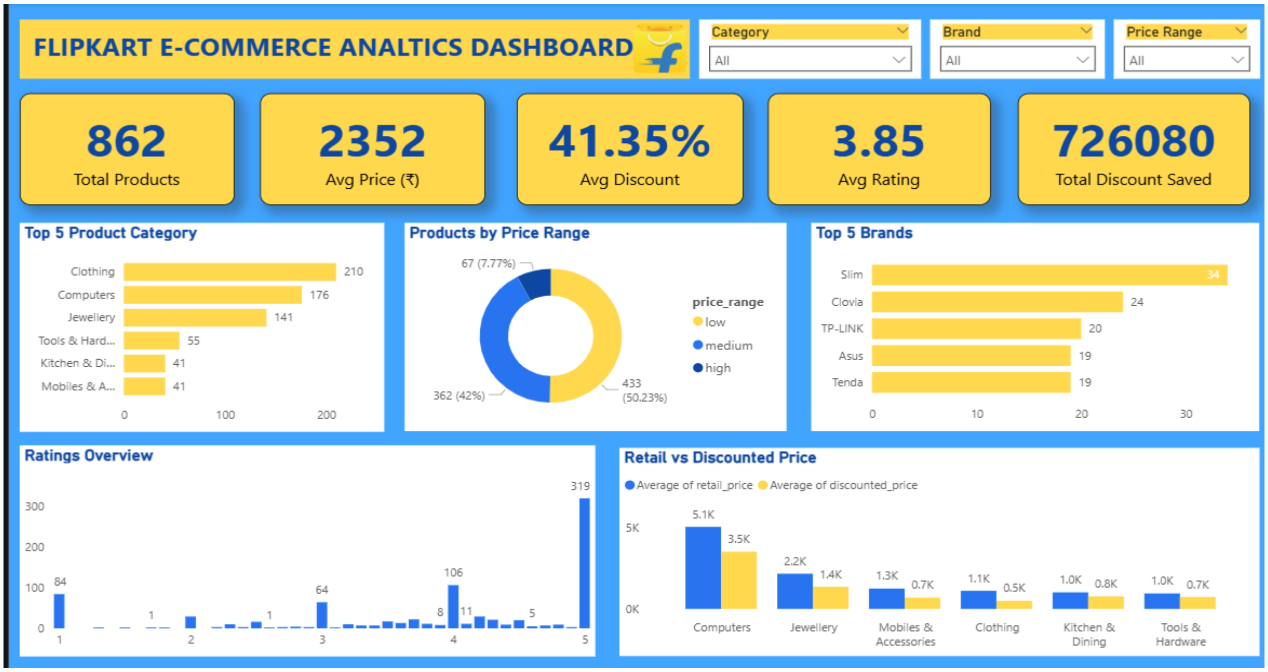
Shows the Top 5 product categories by product count using a clustered bar chart, helping identify dominant product segments.

xlviii. Price and Brand Analysis:

Includes a donut chart for product distribution by price range, a Top 5 Brands bar chart, and a clustered column chart comparing retail price versus discounted price across major categories.

xlix. Ratings Distribution Analysis:

Displays a ratings overview chart showing the distribution of products across different rating values to analyze product quality patterns.



[Figure 5.1: Power BI Dashboard]

5.2 Brief Description of Various Modules

The project comprises the following modules, each serving a distinct analytical purpose:

xlvi. Data Ingestion Module:

Imports the Flipkart raw CSV dataset and performs initial schema validation, attribute inspection, and data loading for analysis.

xlvii. Data Cleaning and Preprocessing Module:

Performs data cleaning using Python and Power Query, including handling missing values, data type conversion, duplicate removal, and feature engineering such as discount percentage, price range, and risk flag generation.

xlviii. Python EDA Module:

Generates statistical summaries and exploratory visualizations using Python libraries to analyze product categories, pricing patterns, discounts, and rating distributions.

xlix. SQL Analysis Module:

Executes business-oriented SQL queries for category analysis, brand analysis, pricing insights, discount evaluation, and product segmentation.

l. Power BI Dashboard Module:

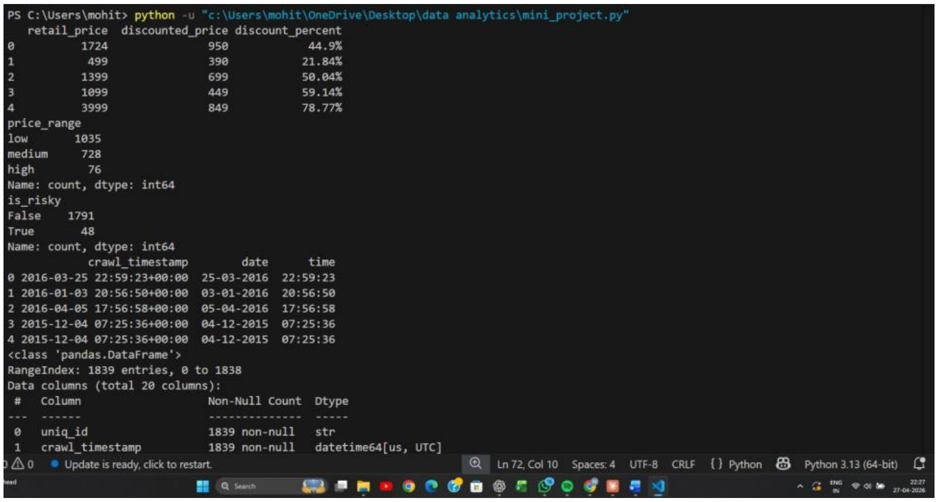
Provides the end-user-facing interactive business intelligence dashboard using KPI cards, slicers, DAX measures, and visualization components for dynamic analysis.

5.3 Snapshots of System with Brief Details

Key system snapshots include:

xlvi. Python Console Output:

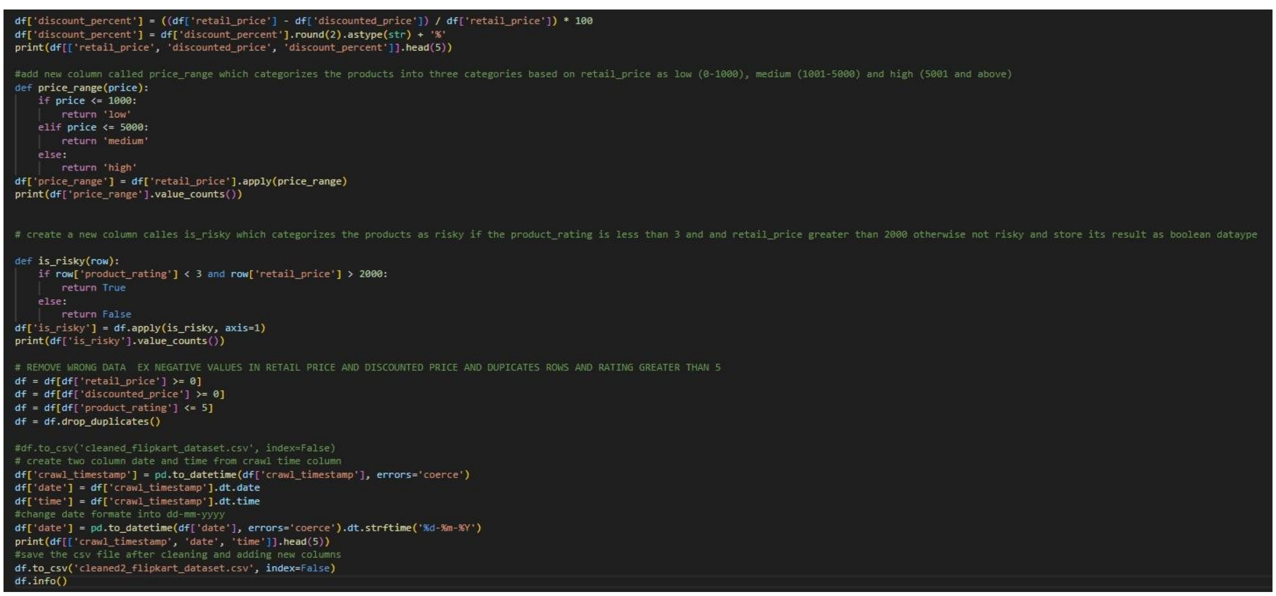
Displays dataset structure, null value handling, data type summary, and cleaned DataFrame output after preprocessing.



```
PS C:\Users\mohit> python -u "c:\Users\mohit\OneDrive\Desktop\data analytics\mini_project.py"
   retail_price  discounted_price discount_percent
0          1724               950            44.9%
1           499               390           21.84%
2          1399               699           50.04%
3          1099               449           59.14%
4          3999               849           78.77%
price_range
low       1035
medium     728
high        76
Name: count, dtype: int64
is_risky
False    1791
True       48
Name: count, dtype: int64
            crawl_timestamp        date      time
0 2016-03-25 22:59:23+00:00  25-03-2016  22:59:23
1 2016-01-03 20:56:50+00:00  03-01-2016  20:56:50
2 2016-04-05 17:56:58+00:00  05-04-2016  17:56:58
3 2015-12-04 07:25:36+00:00  04-12-2015  07:25:36
4 2015-12-04 07:25:36+00:00  04-12-2015  07:25:36
<class 'pandas.DataFrame'>
RangeIndex: 1839 entries, 0 to 1838
Data columns (total 20 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   uniq_id                   1839 non-null   str
 1   crawl_timestamp           1839 non-null   datetime64[us, UTC]
```


xlvii. Exploratory Data Analysis:

Includes visualizations such as product category distribution, pricing distribution, discount analysis, and ratings analysis generated during EDA.



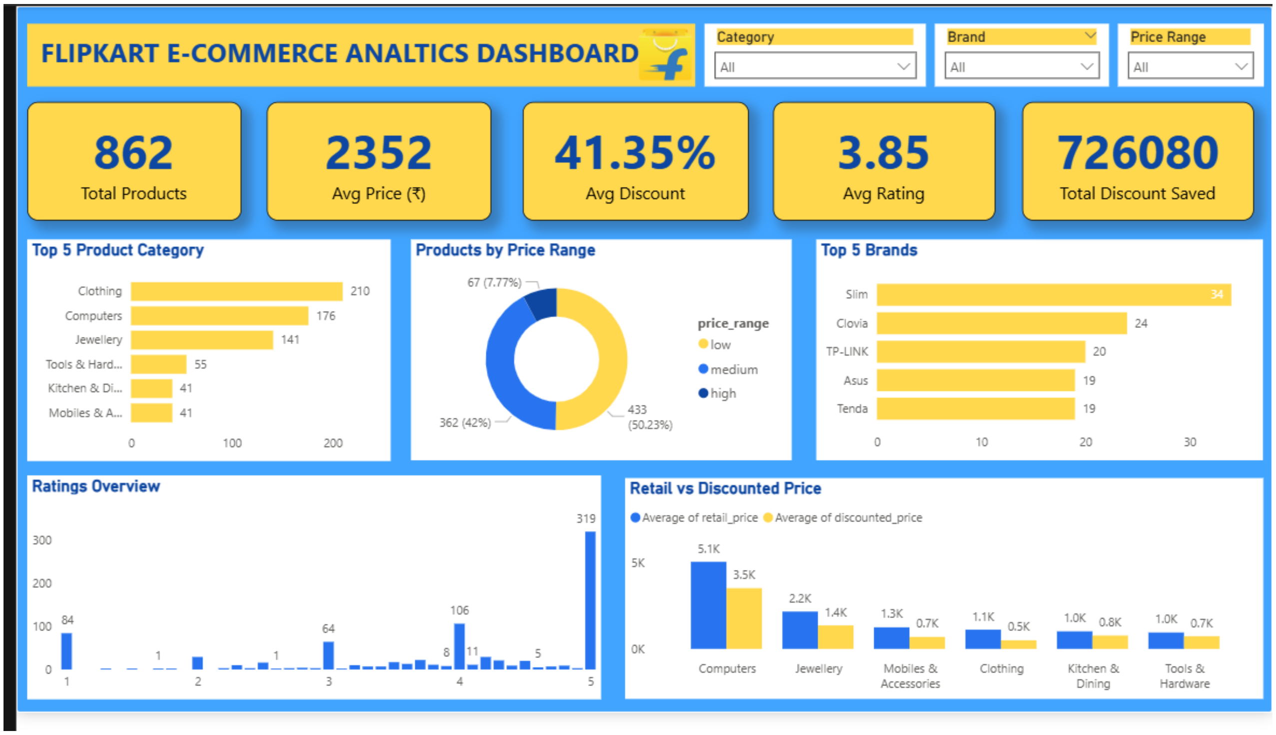
```python
df['discount_percent'] = ((df['retail_price'] - df['discounted_price']) / df['retail_price']) * 100
df['discount_percent'] = df['discount_percent'].round(2).astype(str) + '%'
print(df[['retail_price', 'discounted_price', 'discount_percent']].head(5))

#add new column called price_range which categorizes the products into three categories based on retail_price as low (0-1000), medium (1001-5000) and high (5001 and above)
def price_range(price):
    if price <= 1000:
        return 'low'
    elif price <= 5000:
        return 'medium'
    else:
        return 'high'
df['price_range'] = df['retail_price'].apply(price_range)
print(df['price_range'].value_counts())


# create a new column calles is_risky which categorizes the products as risky if the product_rating is less than 3 and and retail_price greater than 2000 otherwise not risky and store its result as boolean dataype

def is_risky(row):
    if row['product_rating'] < 3 and row['retail_price'] > 2000:
        return True
    else:
        return False
df['is_risky'] = df.apply(is_risky, axis=1)
print(df['is_risky'].value_counts())

# REMOVE WRONG DATA  EX NEGATIVE VALUES IN RETAIL PRICE AND DISCOUNTED PRICE AND DUPICATES ROWS AND RATING GREATER THAN 5
df = df[df['retail_price'] >= 0]
df = df[df['discounted_price'] >= 0]
df = df[df['product_rating'] <= 5]
df = df.drop_duplicates()

#df.to_csv('cleaned_flipkart_dataset.csv', index=False)
# create two column date and time from crawl time column
df['crawl_timestamp'] = pd.to_datetime(df['crawl_timestamp'], errors='coerce')
df['date'] = df['crawl_timestamp'].dt.date
df['time'] = df['crawl_timestamp'].dt.time
#change date formate into dd-mm-yyyy
df['date'] = pd.to_datetime(df['date'], errors='coerce').dt.strftime('%d-%m-%Y')
print(df[['crawl_timestamp', 'date', 'time']].head(5))
#save the csv file after cleaning and adding new columns
df.to_csv('cleaned2_flipkart_dataset.csv', index=False)
df.info()
```

xlviii. SQL Query Results:

Shows outputs of business queries such as top product categories, top brands, average pricing, and discount analysis.

xlix. Power BI Dashboard:

Displays KPI cards, interactive slicers, Top 5 category analysis, brand analysis, price range distribution, ratings overview, and retail vs discounted price comparison.



l. Interactive Dashboard Filtering:

Shows dashboard behavior under category, brand, and price range filtering with dynamically updated KPIs and visuals.

5.4 Back-End Representation

The back-end of this project consists of: (1) a structured product dataset used for analytical processing through SQL queries; (2) Python scripts for ETL operations including data cleaning, transformation, and feature engineering; and (3) a Power BI data model supporting DAX measures, filters, and interactive dashboard reporting. The processed data flows from Python preprocessing to SQL analysis and finally into Power BI for visualization and business intelligence reporting.

Chapter 6: Summary and Conclusions

This project successfully developed an **End-to-End E-Commerce Business Intelligence and Analytics Platform** that transforms raw product data into an interactive Power BI dashboard for generating actionable business insights. The project addresses an important need in e-commerce analytics — converting large product datasets into meaningful insights related to pricing, discounts, brand performance, and product quality for informed decision-making.

The work carried out in this project can be summarized as follows:

xlvi. A complete analytics pipeline was designed and implemented, covering data ingestion, cleaning, exploratory analysis, SQL-based analysis, and interactive dashboard visualization.

xlvii. Python and Power Query were used effectively for data cleaning and preprocessing, including missing value handling, duplicate removal, data transformation, and feature engineering such as discount percentage and price range classification.

xlviii. Python using NumPy, Pandas, and Matplotlib enabled exploratory data analysis, revealing important patterns related to category distribution, pricing trends, discount behavior, and rating distribution.

xlix. SQL queries provided efficient aggregation and business-oriented analysis for category performance, brand analysis, pricing insights, and discount evaluation.

l. A one-page interactive Power BI dashboard was successfully developed to answer business questions related to product categories, pricing strategy, discounts, brand presence, and product segmentation.

Key conclusions drawn from the analysis include:

li. Product inventory is concentrated in a few dominant categories, with categories such as Clothing and related segments showing high product presence.

lii. A large share of products falls within low and medium price ranges, indicating stronger marketplace concentration in affordable products.

liii. Discounting plays a significant role in pricing strategy, with substantial customer savings observed through the Total Discount Saved metric.

liv. Certain brands have stronger product representation, indicating concentration of marketplace offerings among a limited number of brands.

lv. Product ratings show a strong concentration toward higher ratings, suggesting generally positive product quality levels.

Chapter 7: Future Scope

While the current platform successfully addresses core business intelligence requirements, several enhancements are identified for future development:

1. Real-Time Data Integration: Extending the platform to consume live data streams from e-commerce APIs (e.g., Shopify, WooCommerce, Amazon Seller API) using Apache Kafka or Azure Event Hubs, enabling real-time dashboard refresh.Machine Learning Integration: Incorporating predictive analytics models — such as sales forecasting using ARIMA or LSTM, customer churn prediction using Logistic Regression, and product recommendation using collaborative filtering — directly into the Power BI dashboard via Python visuals.
2. Cloud Deployment: Migrating the SQL database to a cloud-based solution such as Azure SQL or Google BigQuery, and publishing the dashboard to Power BI Service for web and mobile access.
3. Automated ETL Pipeline: Building a fully automated ETL pipeline using Apache Airflow or Azure Data Factory to schedule daily data refreshes without manual intervention.
4. Natural Language Querying: Leveraging Power BI's Q&A feature and Azure OpenAI integration to allow business users to query the dashboard using natural language (e.g., 'Show me sales in the West region for Q3 2024').Advanced Customer Analytics: Implementing RFM (Recency, Frequency, Monetary) analysis for deeper customer segmentation and lifetime value (CLV) prediction.
5. Mobile Dashboard Optimization: Designing mobile-optimized Power BI report layouts for use on tablets and smartphones.
6. Data Governance and Security: Implementing row-level security (RLS) in Power BI to ensure that regional managers only see data relevant to their geography.

These enhancements would transform the current academic prototype into a production-ready, enterprise-grade business intelligence solution suitable for real-world e-commerce deployments.

Appendix

Appendix I – Sample Python Code: Data Cleaning

The following Python script illustrates the key data cleaning operations performed using Pandas:

```
import pandas as pd
import numpy as np

df = pd.read_csv('ecommerce_raw.csv')
df.drop_duplicates(inplace=True)
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df['OrderDate'] = pd.to_datetime(df['OrderDate'])
df['ProfitMargin'] = (df['Profit'] / df['Sales']) * 100
df.to_csv('ecommerce_clean.csv', index=False)
```

Appendix II – Sample SQL Queries

```
-- Total Revenue by Category
SELECT Category, ROUND(SUM(Sales),2) AS Total_Revenue,
ROUND(SUM(Profit),2) AS Total_Profit
FROM Orders GROUP BY Category ORDER BY Total_Revenue DESC;

-- Monthly Sales Trend
SELECT strftime('%Y-%m', OrderDate) AS Month,
ROUND(SUM(Sales),2) AS Monthly_Revenue
FROM Orders GROUP BY Month ORDER BY Month;
```

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